SC220-40 / SC220-80 — Job File Guide

This guide describes the job file of **SC220-40 and SC220-80 machines with the MSC interface**. The file layout is **identical to the NR241 job file**; the only difference is the structure of the **operation / macro block** in the `nccode` field.

File fields

The file has the same **30 columns** as NR241:

```
id;ksn;ksnbar;ktn;ktnbar;l;r;code;info;width;height;trolley;box;orientation;reinf;reinfbar;pos;prono;offno;ag
ent;customer;date;nccode;isfix;colorcode;colorinfo;mainprofile;subcust;image;DATA1
```

For field meanings, bar and part sequence, and length and angle notation, see the NR241 / NR242 / NR210 guide. In short:

- Each data row is one part to be cut; rows with the same `ksn` value are cut from the same bar.
- In `ksnbar` and `ktnbar`, the last digit is the decimal: `4710` = **471.0 mm**.
- `l` and `r` are the start and end cutting angles, written as a single number (`45`, `135`).
- If a field is empty, keep its column position; do not shift the following values to the left.

`nccode` — operation / macro field

On SC220 machines each operation is a **12-character** block. If a part has more than one operation, the blocks are written one after another.

```
H | 02 | 01 | 01 | 05000
```

The vertical bars are for explanation only; they are not part of the field.

Part	Character position	Length	Description
`H`	1	1	Block start character
`02`	2–3	2	Tool code
`01`	4–5	2	Operation code
`01`	6–7	2	Accessory code
`05000`	8–12	5	Operation position; the last digit is the decimal: **500.0 mm**

The full block for this example:

```
H02010105000
```

Keep leading zeros: write Tool `02`, not `2`, and position `05000`, not `5000`.

Difference from NR241

	NR241 / NR242 / NR210	SC220-40 / SC220-80
Block length	18 characters	**12 characters**
Tool code	3 digits (`002`)	**2 digits** (`02`)
Operation / Accessory	2 + 2 digits	2 + 2 digits
Position	5 digits, 0.1 mm	5 digits, 0.1 mm
Slot length	5 digits	**None**

The same operation in both formats:

```
NR241 : H00201010500001000
SC220 : H02010105000
```

Do not import an NR-format `nccode` field into an SC220 machine as is; the blocks will be read incorrectly.

Hole example

```
H16010110000
```

Tool **`16`**, Operation **`01`**, Accessory **`01`**; position **1000.0 mm**.

Several operations on one part

```
H02010105000H16010110000
```

Two 12-character blocks, **24 characters** in total. Blocks are split at the `H` character, so `H` must not appear inside a block.

If a block is shorter than 12 characters or contains non-numeric characters, the machine cannot read **any operation** of that part. The operations will be missing even if the cutting lengths are correct.

These codes are examples, not a universal operation code list for every profile and machine. Check what each operation code means on your machine together with Macro Mapping and the profile's operation definitions.

Example file

Download the 30-column SC220 example CSV

Dealer, customer, order, profile and image path values in the example are fictitious. The separator is `;`. Match it to your machine's profile, operation and import settings before production.

Before importing

1. Compare the column headers and the number of fields per row with the NR241 layout.
2. Check bar numbers, part sequence within each bar, bar / part lengths and cutting angles.
3. Check that `nccode` blocks are **12 characters** and the Tool code is written with **2 digits**; keep leading zeros.
4. Make sure the file was not prepared in NR format (18-character blocks).

5. After importing, check the lengths and operation view in the part list.

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